

Loss of Mixed Unitarity after a Positive Time in an Explicit Qutrit Semigroup

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Abstract

We give a fixed unital Lindblad generator on $\mathbb{M}_3$ whose semigroup is mixed unitary at a strictly positive time and is not mixed unitary at a later time. The two times are $s = 2\pi/\sqrt{3601}$ and $t = 3\pi/\sqrt{3601}$. At s , an exact block-evolution identity expresses the channel as a unitary conjugation composed with a semigroup generated by five Hermitian jumps. Compactness then gives membership in the finite convex hull of unitary conjugations. At t , a twisted-swap witness, valid for every complex qutrit unitary, has expectation strictly less than $-456/38125$. The construction and both certificates are analytic.

1 Question, conventions, and statement

A channel on $\mathbb{M}_d = \mathcal{L}(\mathbb{C}^d)$ is *mixed unitary* if it has a finite representation

$$\Psi(X) = \sum_{j=1}^N p_j U_j X U_j^*, \quad p_j \geq 0, \quad \sum_{j=1}^N p_j = 1, \quad (1)$$

where each U_j is unitary. We write $\text{Ad}_U(X) = UXU^*$. Throughout, $*$ denotes adjoint, T denotes transpose in the displayed computational basis, and $\bar{}$ denotes complex conjugation. Matrices in block calculations are arbitrary complex matrices; no Hermiticity restriction is imposed on their entries.

Bhat and Devendra prove eventual mixed unitarity for every dynamical semigroup of unital quantum channels [1, Theorem 4.12]. Their Theorem 5.4 describes an initial interval of nonmembership when the semigroup is not mixed unitary at every time. The subsequent open question asks whether entry into the mixed-unitary set can precede the time after which membership persists. The corresponding QIQC question [2] asks for one norm-continuous unital completely positive trace-preserving (CPTP) semigroup and two times $0 < s < t$ with membership at s and nonmembership at t .

The dissipative part used below is the qutrit construction in [1, Lemma 5.7 and Example 5.11]. The twisted-swap separation method also appears in [1, Lemmas 5.5–5.7]. Here a specified Hamiltonian produces an exactly solvable refocusing time. We include the membership and separation arguments in full.

For a map $\Psi : \mathbb{M}_3 \rightarrow \mathbb{M}_3$, use the normalized Choi convention

$$J(\Psi) = (\text{id} \otimes \Psi)(|\Omega\rangle\langle\Omega|), \quad |\Omega\rangle = \frac{1}{\sqrt{3}} \sum_{j=0}^2 |jj\rangle. \quad (2)$$

In particular, $\text{Tr } J(\Psi) = 1$ for a trace-preserving map. Let $E_{ij} = |i\rangle\langle j|$ and let $F = \sum_{i,j=0}^2 E_{ij} \otimes E_{ji}$ be the swap operator.

Theorem 1. *Set*

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \text{diag}(1, R), \quad H = \text{diag}(30, 0, 0), \quad (3)$$

and define

$$\mathcal{L}(X) = \frac{\text{Tr}(X)I_3 - BX^TB^*}{2} - X - i[H, X]. \quad (4)$$

Then $\Phi_u = e^{u\mathcal{L}}$, $u \geq 0$, is a norm-continuous unital CPTP semigroup. For

$$s = \frac{2\pi}{\sqrt{3601}}, \quad t = \frac{3\pi}{\sqrt{3601}}, \quad (5)$$

Φ_s has a finite mixed-unitary representation and Φ_t does not. More precisely, set

$$C = \text{diag}(-1, R), \quad S_C = (I_3 \otimes C)F(I_3 \otimes C^*), \quad W = S_C + \frac{I_9}{3}. \quad (6)$$

Every mixed-unitary channel Ψ satisfies $\text{Tr}(WJ(\Psi)) \geq 0$, whereas

$$\text{Tr}(WJ(\Phi_t)) = \frac{2 + e^{-3t/2} - 3e^{-t/2} - 4e^{-t}/\sqrt{3601}}{3} < -\frac{456}{38125}. \quad (7)$$

2 A unital Lindblad generator

Define the map

$$T(X) = \frac{\text{Tr}(X)I_3 - BX^TB^*}{2}. \quad (8)$$

Its complete positivity can be checked directly from

$$J(T) = \frac{I_9 - (I_3 \otimes B)F(I_3 \otimes B^*)}{6} \geq 0. \quad (9)$$

Indeed, B is unitary, and the conjugated swap is a Hermitian involution. Direct evaluation gives $T(I_3) = I_3$ and $\text{Tr } T(X) = \text{Tr } X$.

For an explicit Lindblad representation, put

$$L_{ij} = \frac{B(E_{ij} - E_{ji})}{\sqrt{2}}, \quad 0 \leq i < j \leq 2. \quad (10)$$

Expanding the three terms yields

$$T(X) = \sum_{i < j} L_{ij} X L_{ij}^*, \quad \sum_{i < j} L_{ij}^* L_{ij} = I_3, \quad \sum_{i < j} L_{ij} L_{ij}^* = I_3. \quad (11)$$

For example, before multiplication by B , the first identity is $\sum_{i < j} (E_{ij} - E_{ji})X(E_{ij} - E_{ji})^* = \text{Tr}(X)I_3 - X^T$. Consequently

$$\mathcal{L}(X) = -i[H, X] + \sum_{i < j} \left(L_{ij} X L_{ij}^* - \frac{1}{2} \{L_{ij}^* L_{ij}, X\} \right), \quad (12)$$

where $\{A, X\} = AX + XA$.

The semigroup properties also follow without invoking a generator classification. The Poisson expansion

$$e^{u(T-\text{id})} = e^{-u} \sum_{n=0}^{\infty} \frac{u^n}{n!} T^n \quad (13)$$

is a norm limit of unital CPTP maps, as is seen by assigning any truncated tail weight to id . The Hamiltonian part generates $\text{Ad}_{e^{-iuH}}$. The finite-dimensional Lie product formula gives

$$e^{u\mathcal{L}} = \lim_{n \rightarrow \infty} \left(e^{(u/n)(T-\text{id})} \circ \text{Ad}_{e^{-i(u/n)H}} \right)^n. \quad (14)$$

Complete positivity, trace preservation and unitality survive this norm limit. Finally, finite-dimensional matrix exponentiation gives $\Phi_0 = \text{id}$, $\Phi_{u+v} = \Phi_u \circ \Phi_v$, and norm continuity.

3 Exact block evolution and refocusing

Write

$$X = \begin{pmatrix} x & r \\ c & A \end{pmatrix}, \quad r \in \mathbb{C}^{1 \times 2}, \quad c \in \mathbb{C}^{2 \times 1}, \quad A \in \mathbb{M}_2.$$

The elementary identity

$$RA^T R^* = \text{Tr}(A)I_2 - A \quad (15)$$

holds for every complex 2×2 matrix. Substitution into (4) gives all four block equations:

$$x' = \frac{1}{2} \text{Tr}(A) - x, \quad A' = \frac{x}{2} I_2 - \frac{1}{2} A, \quad (16)$$

$$r' = (-1 - 30i)r + \frac{1}{2} c^T R, \quad c' = (-1 + 30i)c - \frac{1}{2} R r^T. \quad (17)$$

Here is the complete solution of the diagonal blocks. Write the initial data as x_0, r_0, c_0, A_0 , let $\tau = x_0 + \text{Tr } A_0$, and put

$$v(u) = e^{-3u/2}, \quad w(u) = e^{-u/2}. \quad (18)$$

Since $x + \text{Tr } A = \tau$ is constant, the solution of (16) is

$$x(u) = \frac{\tau}{3} + \left(x_0 - \frac{\tau}{3} \right) v(u), \quad (19)$$

$$A(u) = w(u) \left(A_0 - \frac{\text{Tr } A_0}{2} I_2 \right) + \frac{\tau - x(u)}{2} I_2. \quad (20)$$

To solve the other blocks, set $q = c^T R$. Equation (17) implies $q' = (-1 + 30i)q - r/2$, so

$$\frac{d}{du} \begin{pmatrix} r \\ q \end{pmatrix} = (-I_2 + M) \begin{pmatrix} r \\ q \end{pmatrix}, \quad M = \begin{pmatrix} -30i & 1/2 \\ -1/2 & 30i \end{pmatrix}. \quad (21)$$

The displayed scalar matrix acts componentwise on the two rows. With $\omega = \sqrt{3601}/2$, one has $M^2 = -\omega^2 I_2$, hence

$$e^{u(-I_2+M)} = e^{-u} \left(\cos(\omega u) I_2 + \frac{\sin(\omega u)}{\omega} M \right). \quad (22)$$

Equivalently, define

$$a(u) = \cos(\omega u) - i \frac{30}{\omega} \sin(\omega u), \quad k(u) = \frac{\sin(\omega u)}{2\omega}. \quad (23)$$

The remaining blocks are

$$r(u) = e^{-u} (a(u)r_0 + k(u)c_0^T R), \quad (24)$$

$$c(u) = e^{-u} (\overline{a(u)}c_0 - k(u)Rr_0^T). \quad (25)$$

The conjugate of $a(u)$ is a scalar coefficient; these formulas impose no relation between r_0 and c_0 .

Consider the auxiliary generator

$$\mathcal{D}_0 \begin{pmatrix} x & r \\ c & A \end{pmatrix} = \begin{pmatrix} \text{Tr}(A)/2 - x & -r \\ -c & xI_2/2 - A/2 \end{pmatrix}. \quad (26)$$

Its diagonal equations agree with (16), while both off-diagonal blocks are multiplied by e^{-u} under $e^{u\mathcal{D}_0}$. At $s = \pi/\omega$, (22) is exactly $-e^{-s}I_2$. Thus, for $Q = \text{diag}(-1, 1, 1)$,

$$\boxed{\Phi_s = \text{Ad}_Q \circ e^{s\mathcal{D}_0}.} \quad (27)$$

This equality holds on all of $\mathbb{M}_3$. It compares two channels at one specified time and does not assume that their generators coincide.

4 Finite mixed-unitary membership at the positive time

For any matrix V , use the dissipator notation

$$\mathcal{D}_V(X) = VXV^* - \frac{1}{2}(V^*VX + XV^*V). \quad (28)$$

Let $P = \text{diag}(0, 1, 1)$. Direct block multiplication gives

$$\mathcal{D}_0 = \sum_{a=1}^2 \left(\mathcal{D}_{E_{0a}/\sqrt{2}} + \mathcal{D}_{E_{a0}/\sqrt{2}} \right) + \mathcal{D}_{P/\sqrt{2}}. \quad (29)$$

The paired terms give the diagonal part in (26) and off-diagonal decay $-3r/4, -3c/4$. The last term contributes the remaining $-r/4, -c/4$ and leaves the diagonal blocks unchanged. Define the Hermitian matrices

$$A_a = \frac{E_{0a} + E_{a0}}{2}, \quad K_a = \frac{i(E_{0a} - E_{a0})}{2}, \quad a = 1, 2. \quad (30)$$

For each a , expansion of the two dissipators shows that

$$\mathcal{D}_{E_{0a}/\sqrt{2}} + \mathcal{D}_{E_{a0}/\sqrt{2}} = \mathcal{D}_{A_a} + \mathcal{D}_{K_a}. \quad (31)$$

Thus $\mathcal{D}_0$ is a sum of five Hermitian-jump generators.

Lemma 1. *If $D = \sum_{j=1}^m \mathcal{D}_{V_j}$ with $V_j = V_j^*$ in a fixed finite dimension, then e^{uD} is mixed unitary in the finite-mixture sense for every $u \geq 0$.*

Proof. For one Hermitian V , $\mathcal{D}_V(X) = -[V, [V, X]]/2$. In an eigenbasis of V , with eigenvalues λ_a , its exponential multiplies the (a, b) matrix entry by $\exp[-u(\lambda_a - \lambda_b)^2/2]$. For $u > 0$ this is the characteristic function of a centered Gaussian of variance u , giving

$$e^{u\mathcal{D}_V}(X) = \frac{1}{\sqrt{2\pi u}} \int_{\mathbb{R}} e^{-z^2/(2u)} e^{-izV} X e^{izV} dz. \quad (32)$$

At $u = 0$ the map is the identity.

The set of unitary conjugations is compact in the finite-dimensional real vector space of Hermiticity-preserving maps. Its convex hull is also compact: by Caratheodory's theorem in real dimension N , every point in the hull uses at most $N + 1$ points, so the hull is a continuous image of a compact set. Approximating the Gaussian integral by finite convex sums and using this closedness shows that $e^{u\mathcal{D}_V}$ lies in that convex hull. In particular, the exact channel has a finite representation of the form (1).

Compositions of finite mixed-unitary maps remain mixed unitary, since $\text{Ad}_U \circ \text{Ad}_V = \text{Ad}_{UV}$. The Lie product formula now gives

$$e^{uD} = \lim_{n \rightarrow \infty} \left(e^{(u/n)\mathcal{D}_{V_1}} \circ \dots \circ e^{(u/n)\mathcal{D}_{V_m}} \right)^n. \quad (33)$$

Every approximant is mixed unitary, and closedness gives the same conclusion for the limit, with a finite convex representation. $\square$

Equations (29)–(31) and Lemma 1 show that $e^{s\mathcal{D}_0}$ is mixed unitary. The exact identity (27) therefore proves mixed unitarity of Φ_s . The finite decomposition is established by existence in the finite-dimensional convex hull; no infinite-mixture enlargement of definition (1) is used.

5 A witness for all complex unitary channels

The Hermitian matrix W in (6) separates the later channel from every mixed-unitary channel. We first establish its inequality globally.

Lemma 2. *For every $U \in \mathbb{U}(3)$, $\text{Tr}(WJ(\text{Ad}_U)) \geq 0$. Consequently the same inequality holds for every mixed-unitary channel.*

Proof. Put $V = C^*U$, which is an arbitrary complex unitary. From (2) and the action of the swap,

$$\text{Tr}(S_C J(\text{Ad}_U)) = \langle \Omega | (I_3 \otimes V^*) F(I_3 \otimes V) | \Omega \rangle = \frac{1}{3} \text{Tr}(V^* V^T). \quad (34)$$

Let $V_+ = (V + V^T)/2$ and $V_- = (V - V^T)/2$. The symmetric and skew-symmetric parts are orthogonal for the complex Frobenius inner product: terms with interchanged indices cancel. Therefore

$$\text{Tr}(V^* V^T) = \|V_+\|_F^2 - \|V_-\|_F^2 = 3 - 2\|V_-\|_F^2. \quad (35)$$

In odd dimension three, $(V_-)^T = -V_-$ implies $\det V_- = -\det V_-$, hence $\text{rank } V_- \leq 2$. Both V and V^T are unitary, so

$$\|V_-\|_{\text{op}} \leq \frac{\|V\|_{\text{op}} + \|V^T\|_{\text{op}}}{2} = 1, \quad \|V_-\|_F^2 \leq 2.$$

Equations (34)–(35) give a lower bound $-1/3$ for the swap expectation. Since $\text{Tr } J(\text{Ad}_U) = 1$, the $I_9/3$ term in W shifts this bound to zero. Linearity proves the result for every finite convex combination (1). $\square$

6 Exact later-time evaluation and a rational negative bound

We evaluate the witness from the full block solution, with no restriction on the tested unitary family. For brevity write $v = v(u)$, $w = w(u)$ and $k = k(u)$. Equations (19)–(20) give

$$[\Phi_u(E_{00})]_{00} = \frac{1+2v}{3}, \quad [\Phi_u(E_{bd})]_{\text{lower}} = wE_{bd}^{(2)} + \delta_{bd}\gamma I_2, \quad b, d \in \{1, 2\}, \quad (36)$$

where $\gamma = (2+v)/6 - w/2$, and $E_{bd}^{(2)}$ is the matrix unit in the lower 2×2 block.

Expanding the normalized Choi matrix and the swap yields the four-index contraction

$$\text{Tr}(S_C J(\Phi_u)) = \frac{1}{3} \sum_{i,j,a,b=0}^2 C_{aj} \overline{C_{bi}} [\Phi_u(E_{ji})]_{ba}. \quad (37)$$

To spell out this calculation, split the sum *inside* its factor $1/3$ into four groups according to whether i, j are zero or in $\{1, 2\}$. They contribute

$$i = j = 0 : \quad \frac{1+2v}{3}, \quad (38)$$

$$i, j \in \{1, 2\} : \quad -2w + 2\gamma = \frac{2+v}{3} - 3w, \quad (39)$$

$$i = 0, j \in \{1, 2\} : \quad 2e^{-u}k, \quad (40)$$

$$i \in \{1, 2\}, j = 0 : \quad 2e^{-u}k. \quad (41)$$

For (39), use $\sum_{i,j=1}^2 R_{ij}R_{ji} = -2$ and $\sum_{a,i=1}^2 R_{ai}^2 = 2$ in (36). For (40), the input E_{j0} has $c_0 = e_j$ and $r_0 = 0$, so (24) gives $[\Phi_u(E_{j0})]_{0a} = e^{-u}kR_{ja}$; the factor $C_{00} = -1$ then gives $-e^{-u}k \text{Tr}(R^2) = 2e^{-u}k$. For (41), equation (25) gives $[\Phi_u(E_{0i})]_{b0} = -e^{-u}kR_{bi}$, and the sum is $e^{-u}k \text{Tr}(R^*R) = 2e^{-u}k$.

Adding the four groups and the identity term in W , with $\text{Tr } J(\Phi_u) = 1$, proves the exact formula

$$\text{Tr}(WJ(\Phi_u)) = \frac{2 + e^{-3u/2} - 3e^{-u/2} + 4e^{-u}k(u)}{3}. \quad (42)$$

At the later time $t = 3\pi/(2\omega)$, one has $\sin(\omega t) = -1$ and $k(t) = -1/\sqrt{3601}$. This gives the equality in (7).

It remains to prove the strict sign using rational bounds. Put $y = e^{-t/2} \in (0, 1)$. Then

$$2 + e^{-3t/2} - 3e^{-t/2} = 2 + y^3 - 3y = (1 - y)^2(y + 2) \leq \frac{3t^2}{4}, \quad (43)$$

using $1 - e^{-t/2} \leq t/2$ and $y + 2 \leq 3$. The elementary estimates $60 < \sqrt{3601} < 61$ and $\pi < 22/7$ imply

$$0 < t < \frac{11}{70} < \frac{4}{25}. \quad (44)$$

Consequently the positive expression in (43) is strictly smaller than $12/625$. On the other hand,

$$\frac{4e^{-t}}{\sqrt{3601}} > \frac{4(1-t)}{61} > \frac{84}{1525}, \quad (45)$$

where $e^{-t} > 1 - t$ and (44) were used. Combining these inequalities gives

$$\text{Tr}(WJ(\Phi_t)) < \frac{1}{3} \left(\frac{12}{625} - \frac{84}{1525} \right) = -\frac{456}{38125} < 0. \quad (46)$$

Lemma 2 now rules out every mixed-unitary representation of Φ_t . Together with (27), Lemma 1, and $0 < s < t$, this completes the proof of Theorem 1.

7 Scope of the construction

The same fixed generator is used at both times. Membership at s follows from an exact equality of linear maps and compactness of the finite convex hull. Nonmembership at t follows from a witness valid on all complex unitary channels and a strict rational inequality. Floating-point reproductions serve only as supplementary error checks and are not premises of this proof.

The result addresses positive-time loss of mixed unitarity. It does not identify the first positive mixed-unitary time or the eventual membership threshold, and it does not characterize all intermediate membership times. In particular, no discreteness assertion about that set is made. The eventual-mixed-unitarity theorem of Bhat and Devendra is consistent with the later failure exhibited here. Priority and broader literature novelty remain separate from the stated analytic claim.

A Reproducible diagnostics and evidence

The supplement implements the generator directly on all nine complex matrix units, using row-major vectorization. Symbolic calculations check the Kraus formula and the five-Hermitian-jump formula on every matrix unit (18 exact matrix identities in total), both Kraus completeness relations, and $M^2 = -3601I_2/4$. Rational arithmetic also checks the final bound $-456/38125$.

Independent numerical exponentiation checks the refocusing identity, the witness formula, trace preservation, unitality, and Choi positivity at 23 times. The refocusing residual was 2.99×10^{-16} , the largest witness-formula residual was 2.16×10^{-16} , and the largest trace/unital residual was 8.01×10^{-16} . The minimum computed Choi eigenvalue was -1.19×10^{-16} , consistent with roundoff at a zero eigenvalue; positivity is established analytically in the main text. These tests use tolerance 10^{-12} .

A further seeded test of 400 complex unitary matrices checks the swap identity. It does not certify the universal witness inequality, which is proved by the rank argument in Lemma 2. At the later time, 100-digit evaluation gives

$$\mathrm{Tr}(WJ(\Phi_t)) = -0.0134293096502358324643\dots,$$

consistent with the strict rational bound. The sign conclusion does not rely on this decimal approximation.

The complete LaTeX source, `check_semigroup.py`, recorded outputs, software versions, code hashes and an evidence manifest are available with the article at

<https://yuxuanzhang1995.github.io/agentric-research/mixed-unitarity-semigroup/>.

The primary-source comparison uses Bhat and Devendra’s 9 September 2026 version [1]. The construction was originally frozen in research round 10; this revision adds a fresh audit and reproducible supplement, and is not a new round-14 discovery.

AI assistance and review status

AI tools were used to develop candidate arguments, write verification code, assist exposition, and conduct internal adversarial reviews. The revised manuscript uses the evidence-tracking guidance of Scientific Agent Skills [3]. These activities are disclosed as AI assistance, not external refereeing or formal proof verification. No independent human expert approval or exhaustive novelty certification is recorded.

This research preprint is available for mathematical scrutiny. Its theorem is supported by the written argument; the diagnostic software checks finite identities and supplies no substitute for that argument.

References

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